Roll No. | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-2)

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Subject Code : BP-202T

M.Code : 74968

Date of Examination : 15-12-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Briefly write about thefollowing:**

   a) What is electromeric effect?

   b) Write two reactions of alkyl halides.

   c) Differentiate between El and E2 reactions.

   d) Write the tautomer of ethyl asetoacetate and functional isomer of butanone.

   e) What is the hybridisation and geometry of ethene?

   f) Give chemical test to distinguish between an acid and an ester.

   g) Explain role of inductive effect in explaining acidity of carboxylic acids.

   h) Give any two methods of preparation of alkenes.

   i) Give the structure and uses of vanillin.

   j) What is Walden inversion?

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SECTION-B

2. Explain in detail the kinetics involved, order of reactivity of alkyl halides and rearrangements of carbocations involved in E1 and E2 reactions.

3. Give the various methods of preparation and reactions of carboxylic acids.

4. **Account for the following:**

   a) Basicity of amines

   b) Qualitative tests to differentiate between various types of amines.

SECTION-C

5. Name the various therapeutically used alcohols. Give their structures and uses.

6. Comment upon the stability of conjugated dienes. Give any two reactions of 1,3- butadiene.

7. Write a.comparati.ve note on Aldol and Crossed aldol condensation reactions.

8. Comment upon the various factors affecting SN1 and SN2 reactions.

9. Discuss IUPAC system of nomenclature of open chain organic compounds upto 10 carbon atoms.

10. Write down the various methods of preparation of various carbonyl compounds.

11. Give an account of halogenation of alkanes.

12. Comment upon hybridisation in alkanes.

13. Explain the various tests carried out to detect the presence of carboxylic acids and carboxamides?

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.